(6) $\lim_{x\to\infty}\frac{\sqrt{x^2-1}}{2x+1}=$

(1) 1 (2) 0 (3) -1 (4) $\frac{1}{2}$

(7) $\lim_{x\to 0}\frac{a^x-b^x}{x}=$

(1) $\log ab$ (2) $\log\left(\frac{a}{b}\right)$ (3) $\log\left(\frac{b}{a}\right)$ (4) $\frac{a}{b}$

(8) $\lim_{x\to 0}\frac{8^x-4^x-2^x+1^x}{x^2}=$

(1) $2\log 2$ (2) $2(\log 2)^2$ (3) $\log 2$ (4) $3\log 2$

(9) If $f(x)=x(-1)^{\left\lfloor\frac{1}{x}\right\rfloor}$, $x\le 0$, then the value of $\lim_{x\to 0} f(x)$ is equal to

(1) -1 (2) 0 (3) 2 (4) 4

(10) $\lim_{x\to 3}\lfloor x\rfloor=$

(1) 2 (2) 3 (3) does not exist (4) 0

(11) Let the function f be defined by $f(x)=\begin{cases}3x & 0\le x\le 1\\ -3x+5 & 1<x\le 2\end{cases}$, then

(1) $\lim_{x\to 1} f(x)=1$ (2) $\lim_{x\to 1} f(x)=3$

(3) $\lim_{x\to 1} f(x)=2$ (4) $\lim_{x\to 1} f(x)$ does not exist

(12) If $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=\lfloor x-3\rfloor+|x-4|$ for $x\in\mathbb{R}$, then $\lim_{x\to 3^-} f(x)$ is equal to

(1) -2 (2) -1 (3) 0 (4) 1

(13) $\lim_{x\to 0}\frac{xe^x-\sin x}{x}$ is

(1) 1 (2) 2 (3) 3 (4) 0

(14) If $\lim_{x\to 0}\frac{\sin px}{\tan 3x}=4$, then the value of p is

(1) 6 (2) 9 (3) 12 (4) 4

(15) $\lim_{\alpha\to\pi/4}\frac{\sin\alpha-\cos\alpha}{\alpha-\frac{\pi}{4}}$ is

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) 1 (4) 2

(16) $\lim_{n\to\infty}\left(\frac{1}{n^2}+\frac{2}{n^2}+\frac{3}{n^2}+\ldots+\frac{n}{n^2}\right)$ is

(1) $\frac{1}{2}$ (2) 0 (3) 1 (4) ∞

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